Related work: completeness of relational operations in LLMs

Daryl Alfaro • reviewed 2026-10-12 • companion to the results paper

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Each entry was checked against the paper's abstract or project page (linked). *Not checked* marks details that were not confirmed. The question asked of each paper: **does it measure completeness**, meaning recall over the full answer set, and how that changes as the answer set grows, when the evidence is available to the model?

Closest prior work

HoloBench (Maekawa, Iso, Bhutani; ICLR 2025). [link] *The closest work.*

- *What it does:* turns database queries into long-context text; rows from SQL databases are written out as sentences and mixed with irrelevant rows. Five query types: aggregation, max/min, join, comparison/filter, ranking.
- *What it varies:* context length (4k–64k tokens), information density, how the information is spread, and query complexity. *Scoring:* an LLM judge (GPT-4o-mini) gives exact, partial or no-match credit.
- *Findings:* the amount of information matters more than context length; aggregation degrades sharply while max/min stays easy; GPT-4o is observed failing to return all relevant items as context grows.
- *Gap left for us:* it does not vary the number of correct items on its own, does not score at the set level (missing vs. wrong vs. made-up), and relies on a model as judge.

LOFT (Lee et al., Google DeepMind; 2024, NAACL Findings 2025). [link] A long-context suite (up to 1M tokens) testing whether long-context models can replace retrieval, RAG and SQL. Models rival dedicated retrieval systems but "still face challenges in areas like compositional reasoning that are required in SQL-like tasks." Not designed around completeness or answer-set size. *Not checked:* which SQL datasets it uses and how multi-row answers are scored.

Oolong (Bertsch, Pratapa, Mitamura, Neubig, Gormley; 2025). [link] Long-context aggregation: classify each chunk, then aggregate (counting, label frequency). Frontier models score under 50% at 128K tokens. Covers scalar answers only, not sets.

LongProc (2025). [link] Long procedural generation, including HTML→TSV extraction scored by row-level F1 (outputs up to 8K tokens; 17 models). Row extraction penalises dropped rows, so it is close to "list every row". But it copies every item rather than selecting items that meet relational conditions, and answer-set size isn't varied on its own.

Galois: Querying Large Language Models with SQL (Saeed, De Cao, Papotti; EDBT 2024). [link] Runs SQL with the LLM as the storage layer, answering from memory, and reports "low precision and recall in the returned data"; its query-planning prompts improve completeness. Overlaps only with our memory-only setting.

Questions with many answers (retrieval and memory)

- **QAMPARI** (Amouyal et al.; GEM 2023). [link] List answers (at least 5, on average 13) spread over many Wikipedia paragraphs; recall/precision/F1, best F1 32.8. Retrieval, reading and knowledge failures are confounded.
- **RoMQA** (Zhong et al.; EMNLP Findings 2023). [link] Many-answer questions from Wikidata constraints, with robustness measured across clusters of related questions; zero- and few-shot LLMs perform close to naive baselines.
- **QUEST** (Malaviya et al.; ACL 2023). [link] 3,357 entity-seeking queries with implicit set operations ("shorebirds that are not sandpipers"); retrievers struggle with negation and conjunction. Closest to our set-operation slice, but a retrieval benchmark over real Wikipedia, not reasoning over data in the prompt.

Long-context retrieval and reasoning

- **Lost in the Middle** (Liu et al.; TACL 2024). [link] Accuracy is U-shaped in the position of one relevant document. Our per-position analysis (results paper, appendix) tests the same idea for many relevant rows and finds no U-shape.
- **RULER** (Hsieh et al.; 2024). [link] Includes multi-value needle-in-a-haystack (retrieve *all* values for a key, 4 in the standard setting) and aggregation tasks: retrieving every item, but with very small answer sets.
- **BABILong** (Kuratov et al.; NeurIPS 2024 D&B). [link] bAbI reasoning tasks, including counting and lists/sets, hidden in long text; models use only 10–20% of their context effectively.
- **Michelangelo** (Google DeepMind; 2024). [link] Latent Structure Queries, including a Latent List task; synthetic and outside training data, like ours.
- **Attention Overflow** (Sileo; 2024). [link] Asked which items are missing from a list, models degrade at around 100 items and suggest items already listed.
- **The Position Curse** (Zhang et al.; May 2026). [link] Models fail to locate the last few items in short lists; backward retrieval lags forward retrieval.
- **List Counting Failures Are Not One Phenomenon** (Ait Hou et al.; Sep 2026). [link] Counting items in context fails in model-specific ways (e.g. Llama systematically undercounts), related to our COUNT undercounting.

Tables and SQL

- **Table QA** (WikiTableQuestions, TabFact, TableBench, DataBench): QA over tables in the prompt; most answers are single values or true/false (DataBench includes list answers). None vary answer-set size.
- **Text-to-SQL** (Spider, BIRD): the model writes SQL and a database executes it, so completeness comes free. Our benchmark deliberately asks the model to *be* the query engine.
- **TAG / TAG-Bench** (Biswal et al.; CIDR 2025). [link] Questions over databases needing world knowledge or semantic reasoning; standard methods answer at most 20% correctly.
- **LOTUS, semantic operators** (Patel et al.; VLDB 2025). [link] LLM-driven filter, join and aggregate operators with accuracy guarantees: the systems view, which engineers completeness instead of measuring the raw model.

Where this benchmark fits

	Evidence	Operators	Answer size controlled	Set-level error types	Ground truth
HoloBench	in context	5 types	no (density, length)	no (LLM judge, partial credit)	SQL
LOFT (SQL)	in context	SQL-like	no	not checked	SQL
Oolong	in context	aggregation	no	no (scalar answers)	synthetic labels
LongProc (HTML→TSV)	in context	extraction	output length	row-level F1	deterministic
QAMPARI / RoMQA / QUEST	retrieval, memory	set ops (QUEST)	no	recall / precision	Wikipedia, Wikidata
Galois	memory	SQL	no	precision / recall	knowledge base
RULER MV-NIAH	in context	retrieve-all	4 values	no	synthetic
This work	**in context, memory, retrieval**	**7 types**	**yes, 5–100** (partly confounded with table size; adjusted in analysis)	**missing, near-miss, wrong, made-up, duplicate**	**executed SQL**

What is new here.

1. **The number of correct items is a controlled variable.** Prior work varies context length or information density; HoloBench comes closest but doesn't vary answer-set size on its own. In the current design, answer size is partly confounded with table size; the analysis separates the two by regression, and a fully crossed design is a recommended next step.
2. **Set-level errors against exact ground truth, with no LLM judge.** This is what shows that failures are omissions rather than hallucinations, and catches a model inventing items (991 made-up titles from MiMo-V2.6 Flash on a single question).
3. **One framework for seven operation types and three settings,** so in-context completeness can be compared directly with memory and retrieval.
4. **Reasoning cost.** Models that stay complete reason far longer on larger answer sets (a median of about 9,000 reasoning tokens on 100-item answers). We found no prior benchmark that reports this cost.

Novelty risk: medium. HoloBench, Oolong and LongProc together cover nearby ground: database operations written out as text, aggregation, and row extraction. The case for this benchmark rests on points 1–4, especially the answer-size effect and the error taxonomy. HoloBench should be the main baseline, compared against directly.